

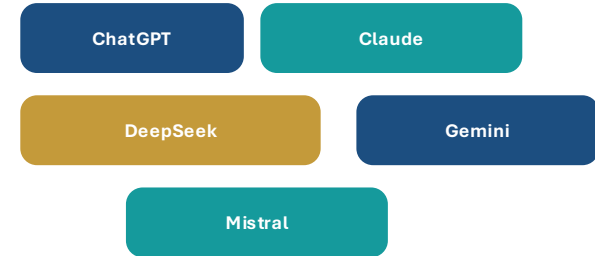
Do Large Language Models Have a Writing Style?

A Stylometric Comparison of LLM-Generated Texts

Research question

Can stylometric features distinguish texts generated by different LLM families?

Viola Awor · Natural Language Processing



LLMs often produce fluent, semantically similar answers to the same prompt.

The open question is whether their writing habits remain measurably different.

- sentence structure
- punctuation habits
- word choice and lexical diversity
- readability and function-word usage
- discourse and AI-style markers

Core distinction

Semantic similarity

what the text is about

Stylometry

how the text is written

Can stylometric features distinguish texts generated by different LLM families?

- Build a balanced corpus of LLM-generated texts.
- Extract measurable stylometric features.
- Test feature differences across model families.
- Compare semantic clustering with stylometric evidence.
- Train classifiers to predict model family.
- Critically discuss effectiveness and limitations.

Item	Value
Generated texts	1,000
Model families	5
Texts per model	200
Genres	4
Texts per genre	250
Texts per model per genre	50
Stylometric features	43
Prompt groups	200

Genres

- Narrative
- Descriptive
- Argumentative
- Dialogue

Why balance matters

Equal model and genre coverage reduces class-size and genre imbalance as sources of bias.



Formal task

Represent each generated text as a 43-dimensional stylometric vector, then predict model family.

$$\phi(x_i) \in \mathbb{R}^{43} \rightarrow \{\text{ChatGPT, Claude, DeepSeek, Gemini, Mistral}\}$$

Length & sentence structure

- word count
- sentence count
- average sentence length
- long-sentence ratio

Punctuation

- comma count
- period count
- dash count
- punctuation per 100 words

Lexical diversity

- type-token ratio
- Yule's K
- Honore's R
- MATTR

Markers & function words

- function-word ratio
- transition markers
- hedging markers
- AI-style markers

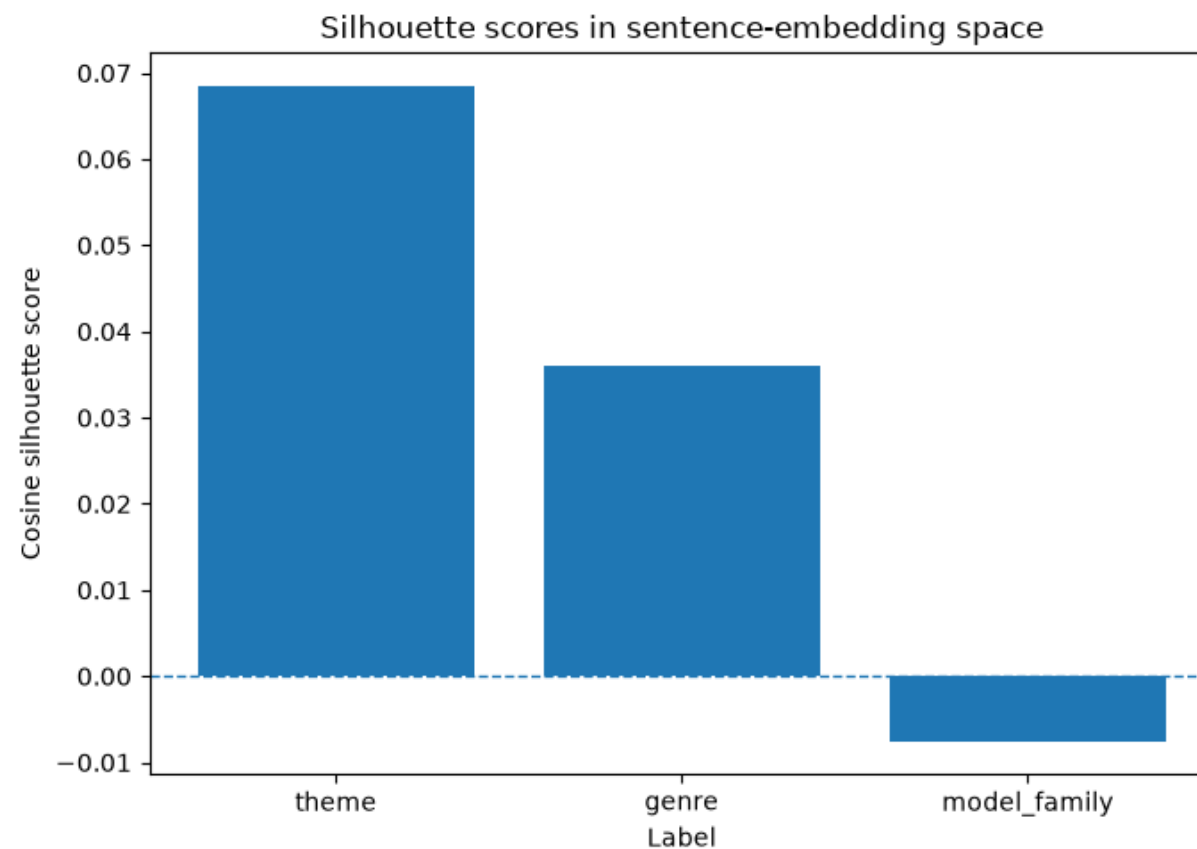
Readability

- Flesch reading ease
- Gunning Fog index

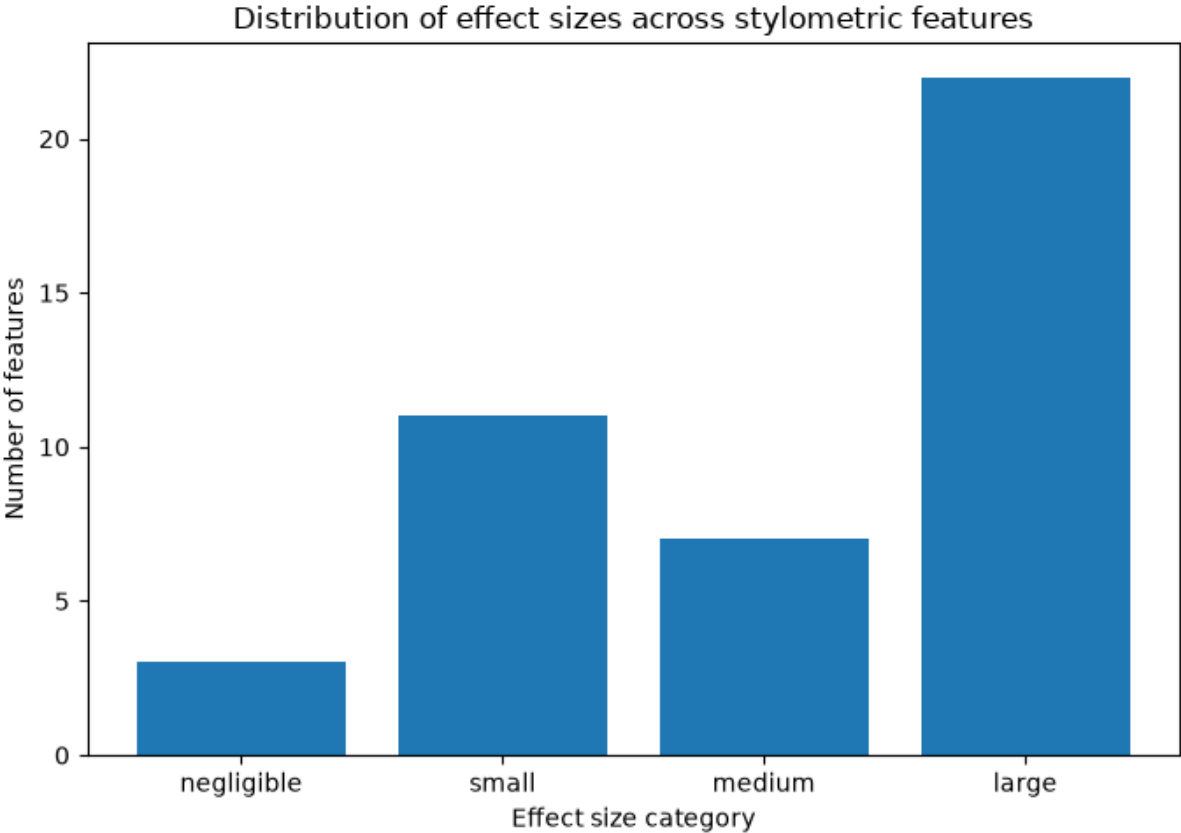
Label	Cosine silhouette
Theme	0.0685
Genre	0.0361
Model family	-0.0075

Interpretation

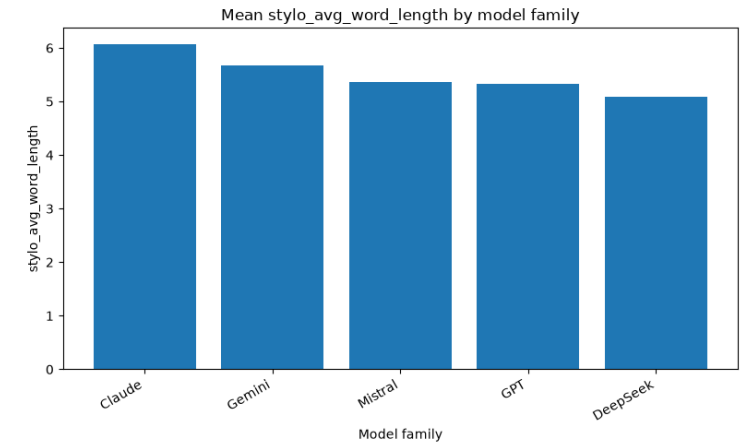
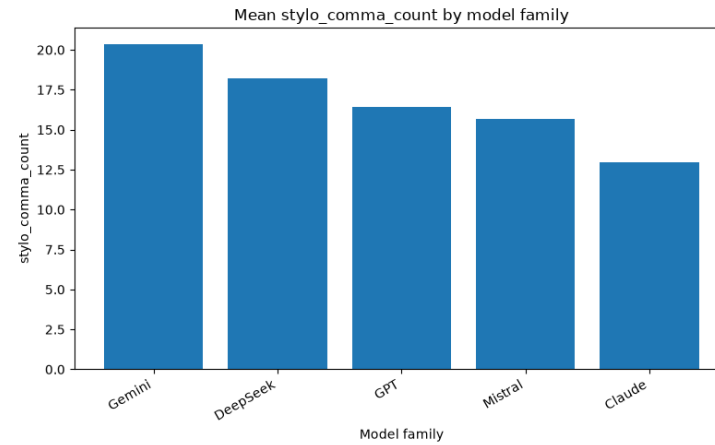
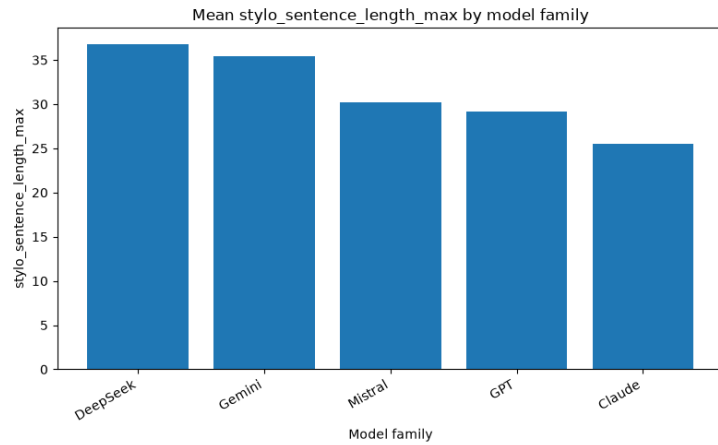
- Theme has the strongest semantic signal, but it is still weak.
- Model family is slightly negative.
- Sentence embeddings capture topic/theme more than model identity.



Result	Value
Features tested	43
Significant after FDR	41
Large effects	22
Medium effects	7
Small effects	11
Negligible effects	3



Most stylometric features differ significantly across model families, and many differences are practically meaningful.



Patterns observed in the project

- DeepSeek: longer and more expansive sentence structures.
- Gemini: higher comma count and comma density.
- Claude: higher average word length and lexical density.
- ChatGPT: higher function-word ratio.

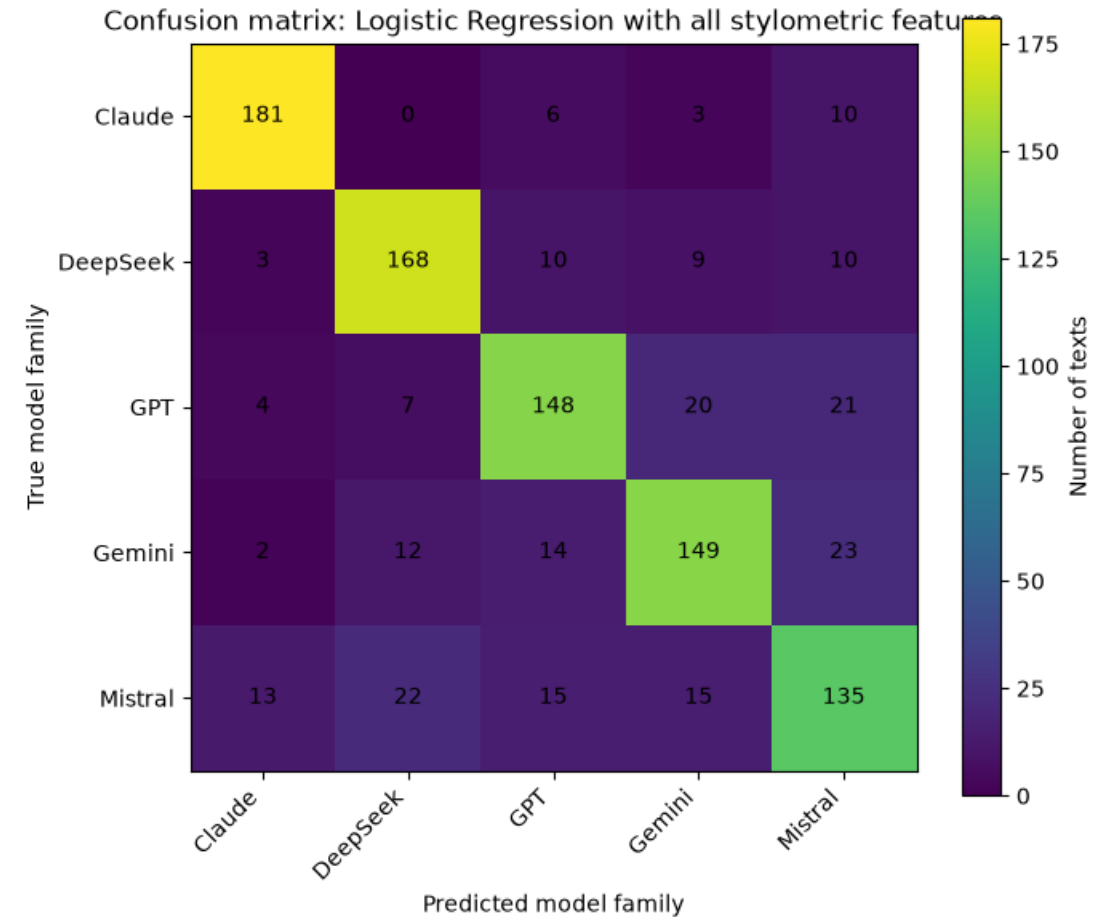
Classification: predicting the generating model family

Supervised evaluation

Classifier	Feature set	Accuracy	Macro F1
Logistic Regression	All stylometric	0.781	0.780
Random Forest	All stylometric	0.779	0.776
Random Forest	Structure/format	0.745	0.742
Logistic Regression	Content-oriented	0.602	0.599

Random baseline = 0.20

Prompt-aware GroupKFold prevents the same prompt ID from appearing in both train and test folds.



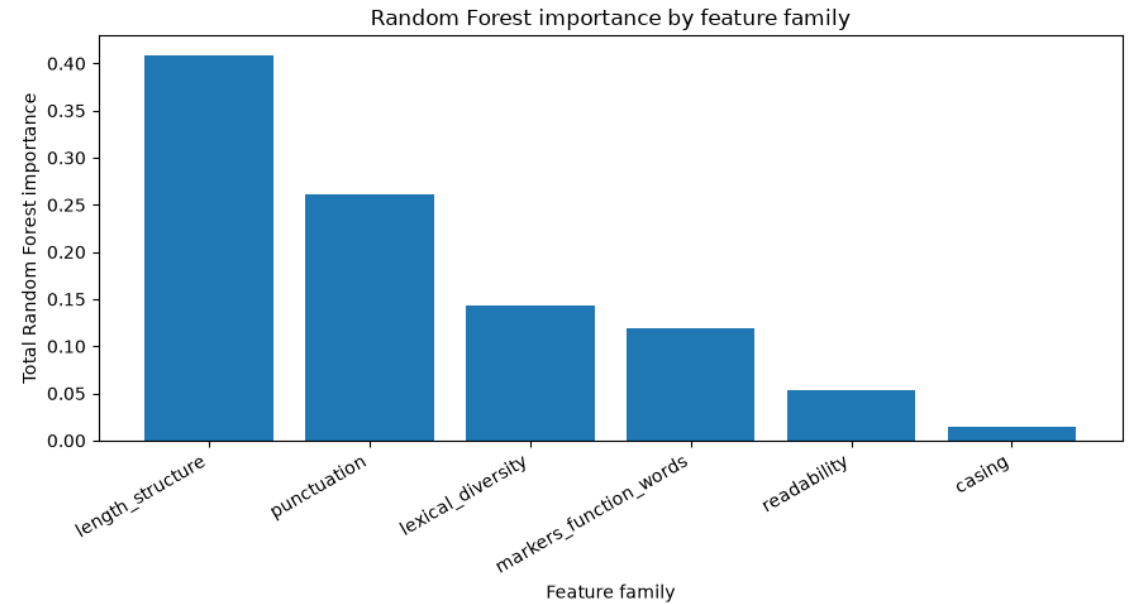
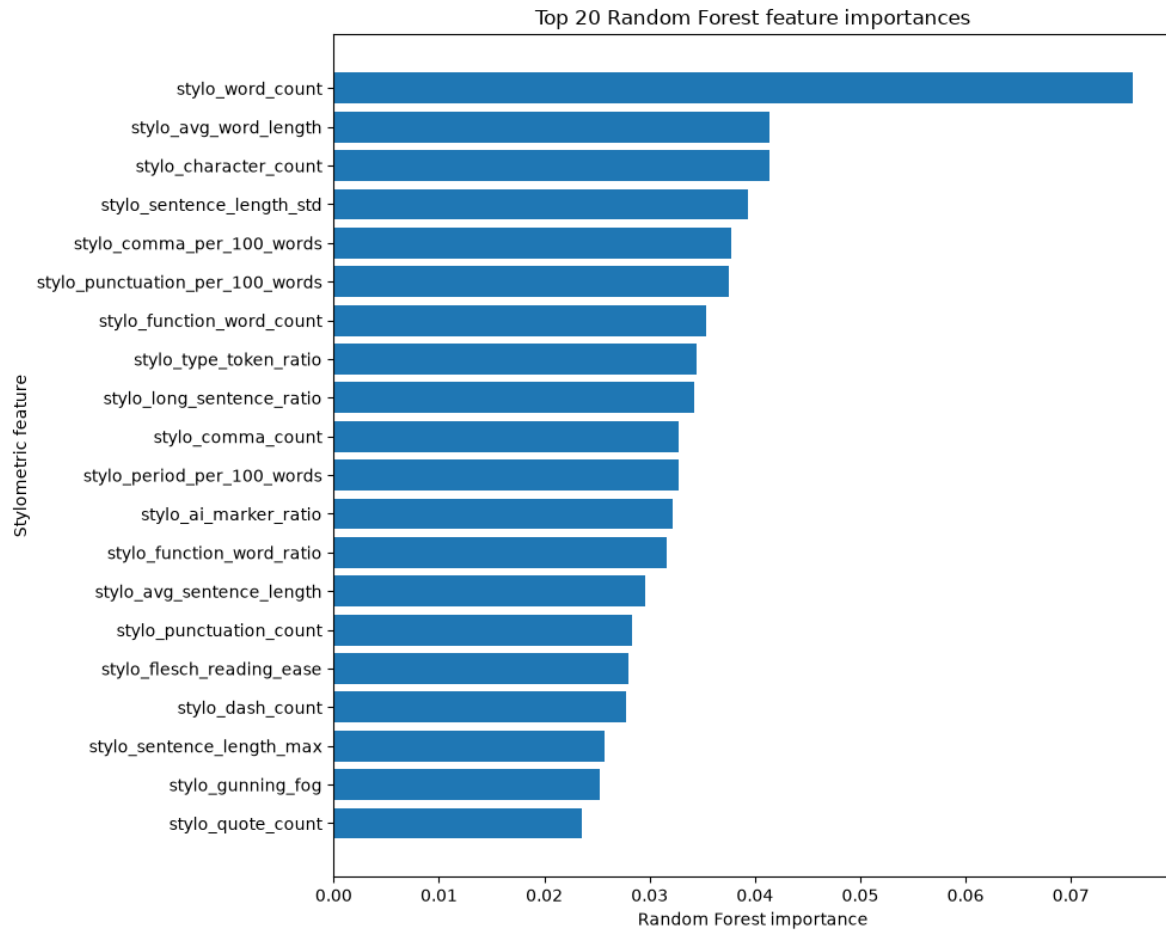
Why prompt-aware splitting matters

- All models answered the same prompt set.
- A random split could leak prompt-specific topic patterns.
- GroupKFold by prompt_id forces generalization to unseen prompts.
- This makes the result a stronger test of writing style.

Evaluation metrics

- Accuracy
- Macro precision
- Macro recall
- Macro F1
- Weighted F1

Feature importance: the fingerprint is distributed



Key point: there is no single “magic” feature; model identity is distributed across several stylistic dimensions.

Interactive dashboard sections

- Dataset summary
- Stylometric feature explorer
- Semantic maps
- Statistical-test results
- Classification performance
- Feature importance and conclusion

Repository organization

- src/: reusable modules
- scripts/: reproducible pipelines
- notebooks/: interpretation
- outputs/: saved tables and figures
- app/: Streamlit dashboard

```
streamlit run app/streamlit_dashboard.py
```

- Corpus is controlled but limited to 1,000 texts.
- Only five model families are included.
- Results may change with different prompts, domains, or languages.
- Model versions and decoding settings can change over time.
- Stylometric features are interpretable but not exhaustive.

The result does not imply perfect attribution in all settings; it shows strong evidence under a controlled experimental design.

Final answer: Yes.

Stylometric features can distinguish texts generated by different LLM families.

Evidence	Main result
Semantic maps	weak model-family clustering
Statistical tests	41 / 43 features significant
Post-hoc tests	86 / 100 pairwise tests significant
Classification	78.1% accuracy vs. 20% baseline

Future work

- more model families and versions
- multilingual prompts
- human-written comparison set
- paraphrasing robustness
- style drift over model updates

LLMs generated semantically similar answers, but they did not write in identical styles.